

How to Get Rejected Before You Submit

Team SAIL

Jihun Yun Kyuhyun Ro Yuchan Lee

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Abstract

The Area Chair’s meta-review is the pivotal artifact of peer review: it fixes the paper’s outcome while auditing whether the author-reviewer discussion actually resolved the raised concerns. **SAIL with Ralph** imitates that dual role end-to-end: a submission receives three independent reviewer reviews, the author rebuts in a threaded discussion with hunk-level AI-drafted revisions, and an AC judge synthesizes the reviews *plus the whole discussion* into a meta-review, a calibrated selectivity score, and an accept/reject decision — revealed, as at the real venue, only at finalization. Built through a sequence of measured calibrations (era and topic conditioning, anti-shortcut decision supervision, rating-scale masking, length conditioning), the decision scorer reaches **89.8% accuracy (AUC 0.965)** on the full 3,767-paper held-out set, and every measurement points the same way: the decisive signal lives in the review-and-rebuttal record, not the manuscript. The judge beats a strong mean-rating rule by 10 accuracy points exactly on borderline papers, reading the discussion adds 3.7 more, deleting a single review swings the decision margin $\sim 30\text{--}50\times$ more than rewriting the abstract, and substituting reviews regenerated from the paper body costs 8 points. Because ICML publishes no meta-reviews, we train synthesis on ICLR (24,673 pairs) and *reverse-generate* a 9,598-paper virtual meta-review corpus for ICML, cleansed of review-history references so each reads as a standalone assessment of the final manuscript; verifying that reinjecting this corpus into training degrades the judge is one of six rejected approaches whose common lesson — behavior lives in the training distribution, not the prompt — drove every calibration above. A sigmoid score head over frozen judge embeddings ranks papers continuously (Spearman 0.872); best-paper winners sit at the top of the embedding space (median 98.7th percentile, a top-10% cut identifies 89%). The agent runs deployed: one GPU serves all heads behind a public review-loop API, a web client, and a terminal reviewer, and the full pipeline re-runs from raw data in about three hours on four H100s.

1 Introduction

Peer review is the quality gate of machine-learning research, but its feedback arrives months after the decisions it could have improved. Of all its artifacts the Area Chair’s meta-review matters most: it simultaneously decides the paper’s fate and audits the author-reviewer discussion — whether concerns were resolved, dodged, or left standing. Our goal is a *review agent*

that imitates exactly that role: a system an author can submit a draft to and receive AC-level feedback — specific reviewer criticisms, a rebuttal loop whose quality the judge actually reads, and a final meta-review with a calibrated score — before the real deadline.

Two design principles distinguish this work. **Venue fidelity**: the agent replicates the ICML process structurally (independent parallel reviews, a discussion phase that the meta-review actually reads, scores that exist only after finalization), because our models are trained on artifacts of exactly that process. **Measured calibration**: every capability claim below is backed by held-out measurement, and every shortcut we could imagine (rating-copying, title memorization, year-to-accept priors, self-training collapse) has an explicit ablation. Each measurement fed directly into the next correction, and the accumulation of those corrections is the system’s current accuracy; rejected approaches are reported alongside what they fixed.

Relation to prior work. Acceptance prediction from review text goes back to PeerRead-style classifiers; meta-review generation has been studied as summarization (e.g., MReD); and agentic reviewing appears inside autonomous-research systems (AI-Scientist-style reviewer loops) and venue-scale LLM-reviewer studies. Our contribution is orthogonal to any single component: a *process-faithful composition* — decisions and text from one judge trained with anti-shortcut supervision on the discussion-bearing input a real AC sees, a separately calibrated selectivity head, and a rebuttal loop whose reviewer reactions are grounded in each reviewer’s own review — evaluated against baselines those literatures rarely report (a strong rating rule, and the venue’s own borderline regime).

2 System Overview

A submission enters a **review cycle** (Fig. 1): (1) three reviewer agents read the manuscript independently and produce structured reviews (calibrated 1–10 rating, confidence, soundness/presentation/contribution facets, strengths, and per-issue comments with severity and section anchors); (2) the author replies in a threaded discussion — addressed reviewers respond in character, grounded in their own review, and judge whether concerns are *resolved, partially addressed, or standing*;

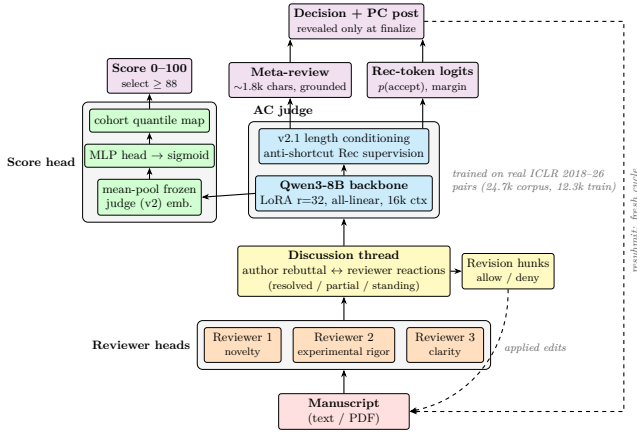


Figure 1: One review cycle. Three reviewer agents produce structured reviews; the discussion thread (rebuttal, reviewer reactions, approved revision hunks) feeds the AC judge together with the reviews. The judge emits the meta-review and the recommendation-token logits; the frozen backbone also feeds the selectivity score head. Score and decision exist only at finalize, as at the venue.

(3) an AI revision agent proposes *hunk-level* manuscript edits (exact-substring anchored, no fabricated results), each of which the author allows or denies; (4) **finalize**: the AC model synthesizes reviews plus the entire discussion into a meta-review, a selectivity score (0–100), an accept/reject decision, and a Program-Chairs decision post; (5) **resubmit** starts a fresh cycle on the revised manuscript. Long operations run as asynchronous jobs streaming progress events to the UI.

The model layer is served on VESSL (vLLM multi-LoRA + a score-head service on one GPU); a lightweight GCE backend owns the public contract-v2 API the web frontend consumes, plus a standalone `/api/score` endpoint.

Model portfolio. The system is a portfolio of specialized heads rather than a version ladder: *v1* (reviews-only, 8k ctx) is a cheap decision scorer; *v2* (discussion-reading, 16k ctx) is the rebuttal judge and the backbone of the product loop; *v2.1* adds length conditioning and is the default meta-review generator; the *score head* rides on frozen *v2* embeddings; reviewer heads are frontier-LLM agents with calibrated-rating prompts (a fine-tuned replacement is staged, §4). All LoRA adapters serve simultaneously from one vLLM instance and are hot-swappable per request.

3 Models and Training

Data. 24,673 (reviews → meta-review) pairs from OpenReview (ICLR 2018–2026, UAI 2024), adversarially audited: resubmission-title leakage between train and test removed, decision-marker text stripped from targets, zero cross-contamination of test titles. ICLR 2023 (3,767 papers, rejects included) is the de-

cision benchmark; ICLR 2026 (accept-only) tests style transfer without contamination. Trends and the field’s knowledge level shift constantly, so a paper accepted years ago can read as stale today — a corpus spanning 2018–2026 would otherwise teach yesterday’s acceptance bar as today’s. We correct for this by conditioning every training example on its venue and year, so the judge calibrates to each era’s norms instead of averaging across them; on top of era conditioning, papers are further classified by topic and category and the calibration is corrected within category, so a fading topic’s older accepts do not inflate the current bar for that area.

AC judge (v2/v2.1). Qwen3-8B with LoRA ($r=32$, all-linear), 16k context, packing with flash attention, and assistant-only loss via a custom ChatML template (the stock Qwen3 template silently masks zero tokens). Inputs are title, venue, abstract, full reviewer reviews, and the pre-meta-review discussion thread. *Anti-shortcut decision supervision*: the explicit “Recommendation:” line is trained only on eras with balanced accept/reject labels, and its presence is controlled by an explicit instruction rather than venue-year signatures, severing the year→accept prior. A continuous accept probability is read from the recommendation-token logits at no extra training cost; the unsaturated logit margin backs a quantile-calibrated fallback score.

Length conditioning (v2.1). *v2* generated meta-reviews at half the length of real ACs (median 760 vs. 1,368 chars) because target length was prompt-unexplained variance; inference-time length requests failed measurably (55% ignored, 31% repetition collapse). *v2.1* makes length an explained variable: every training prompt carries “*Aim for roughly N characters*” with N the actual target length; sub-500-char skeleton targets are dropped; the serving-side anti-hallucination guard for missing discussions is injected at training time so it is in-distribution.

Selectivity score head. A 2-layer MLP over mean-pooled frozen (backbone+LoRA) embeddings regresses a continuous selectivity target (rating- and tier-monotone) through a sigmoid, then maps through cohort quantiles to the product’s 0–100 scale. Training only the head costs a few dollars.

Reverse-generating ICML meta-reviews. ICML does not publish meta-reviews, so the target venue offers no supervision. We train synthesis on ICLR, then run the judge over all 9,598 ICML-accept review sets to reverse-generate a virtual meta-review corpus. Transfer required one fix: ICML’s 1–5 rating scale collapses decisions when read as ICLR’s 1–10 (2/20 accept on a probe), so ratings are masked at generation (16/20 restored). Because ICML publishes only the *final* camera-ready manuscript, generated meta-reviews referencing pre-rebuttal history would be stale feedback; a cleansing pass rewrites the 63% of outputs carrying such references so each reads as a standalone assessment of the final paper. QC: 98% accept-consistency, tier-monotone

quality, and on 35 real resubmission pairs the criticisms align with the authors’ actual next-version changes at 0.70 (real ACs: 0.65). The corpus is an author-feedback artifact — §4 shows why it must not be recycled into training.

4 Evaluation

Methodology. We maintain a claims-to-evidence verification ladder: every headline capability names the experiment that would falsify it, and a 13-scenario failure analysis (rating-shortcut, title memorization, no-op training, discussion hallucination, ...) is probed by mandatory ablations in every evaluation run. Decision metrics are computed on the *full* labeled test set (3,767 papers) — never head-sliced samples; generation metrics use stratified seeded samples. Text quality uses a two-layer LLM judge: gold-point recall against the real AC’s meta-review and per-claim groundedness against the input reviews.

	Acc.	Borderline	AUC
<i>Protocol A — full test set (3,767), logit read-out</i>			
Mean-rating rule (≥ 5.5)	84.7	69.7	—
v1 (reviews only)	89.8	80.1	0.965
v2.1 (+discussion, length cond.)	89.1	78.8	0.965
<i>Protocol B — stratified 100-paper generation samples ($CI \approx \pm 8pt$)</i>			
Rule / Opus zero-shot (sample S_1)	90.0 / 80.0	80.4 / 60.9	—
Rule / v2 generation (sample S_2)	83.5 / 88.2	— / 76.9	—

Table 1: Accept/reject accuracy (%). Borderline = mean rating 5–6.5, the regime where the rule fails. The two protocols are not mutually comparable and are separated accordingly: cross-protocol deltas are never claimed, and the two generation samples (S_1 , S_2 ; overlap 2 papers) are compared only against the rule *on the same sample*. Note v1 vs. v2.1 are different recipes, not an ablation: the discussion effect (+3.7pt borderline) is measured *within* one model (with/without discussion), and v2.1 is adopted for text quality (recall 0.58 vs. 0.50), not decision accuracy.

What decides acceptance. Three measurements converge on the same conclusion: the review-and-rebuttal record outweighs the manuscript body as decision evidence. (i) The judge beats the mean-rating rule by +10pt exactly where ratings are uninformative (borderline band) — review *prose* carries signal beyond scores; ablations reject the shortcuts (rating-masked 73.4%, title-masked unchanged). (ii) Reading the discussion adds +3.7pt on borderline papers and cuts ungrounded claims by 26% — the rebuttal record is decision-relevant, exactly as the AC role prescribes. (iii) Feeding the judge frontier-LLM reviews regenerated from the paper body instead of the real reviews costs 8pt (0.87→0.79) — the reviewer-written record contains judgment that even a strong model cannot reconstruct from the manuscript alone. (iv) Input-sensitivity probes sharpen the ratio: deleting a *single* review swings the de-

cision margin by 12.4 logits — a full accept↔reject flip — while rewriting the abstract moves it by at most 0.4, a $\sim 30\text{--}50\times$ gap; injecting an author rebuttal turn alone shifts only -0.25 , consistent with the judge discounting unverified author claims (small- n probes). Zero-shot frontier LLMs, reading the same reviews, *lose* to the rating rule.

Meta-review text. An LLM judge measures gold-point recall against the real AC meta-review and claim groundedness against the input reviews. v2.1 (in-band) reaches recall 0.58 vs. v2’s 0.50 at $2.4\times$ the information density of a zero-shot essay baseline, with groundedness 0.88; discussion inputs cut ungrounded claims per meta-review by 26%.

Score head. Spearman 0.872 on test-2023. Best-paper winners concentrate at the top of the embedding-derived score: all nine rank above the 88th percentile (median 98.7), and a top-10% score cut identifies 89% of them without any award supervision in training. Honest transfer caveat: within an accept-only cohort (2026), range-restricted Spearman is 0.418 — fine-ranking *among* accepted papers remains open.

Value hypothesis (C5). On 35 real ICLR resubmission pairs, criticisms generated from the v1 draft align with what authors actually changed in the next version at 0.70, versus 0.65 for the real AC’s meta-review — first direct evidence the feedback is actionable, though $n=35$ and an LLM-judged alignment metric make this a preliminary signal rather than a confirmed effect.

Paper→review stage. With real reviews as input, the judge reaches 0.87 on an unseen 50-paper full-text benchmark; with synthetic frontier-LLM reviews (calibrated-rating prompt) it reaches 0.79 — a true synthetic-review gap of $-8pt$, far smaller than the $-35pt$ first measured before contamination and rating-channel confounds were separated. Errors skew conservative (reject recall 0.89), which is the safe direction for feedback.

Rejected approach	Result	Mechanism
Virtual-data augment	$-4\text{--}9pt$, $2\text{--}4\times$ runaway	self-training collapse
LLM cleansing of it	worse than raw ($-5pt$)	style homogenization
Retrieval few-shot	$-5pt$	stale review anchoring
Reviewer personas	rej-recall 0.89→0.48	operating point only
Length via prompt	55% ignored, 31% loop	out-of-distribution ask
Accept-only reviewer FT	rating AUC 0.482	never saw a bad paper

Table 2: The calibration record: approaches rejected by measurement, each feeding a design decision (synthetic corpus scoped to feedback-only, training-side length conditioning, calibrated-rating reviewer prompts, reject-paper injection with years stripped to block an “old-year→harsh” shortcut).

The pattern across all six: *behavior is set by the training distribution, not by prompts*. The fixes that worked — length conditioning, decision-supervision policy, era and topic conditioning, reject-paper injection — all rewrote the distribution, and the judge’s current accuracy (89.8 full-set, 80.1 borderline, AUC 0.965) is the

accumulation of exactly those corrections.

5 Serving and Deployment

One H100/L40S serves the entire model layer: vLLM hosts the shared Qwen3-8B backbone with all LoRA adapters (memory fraction tuned so the score-head service — a second backbone instance with the regression head — coexists on the same GPU), and a FastAPI wrapper owns prompt construction so clients never build prompts. The wrapper injects the v2.1 length line and, for discussion-free inputs, an anti-hallucination guard matching the training distribution; a runaway guard repairs the 15% repetition tail by trimming to the last full sentence and closing with the independently logit-probed recommendation, so no client ever sees a degenerate meta-review. Scores resolve through a fallback ladder: trained head $\rightarrow$ quantile-mapped logit margin $\rightarrow$ raw probability. The GCE backend exposes the twelve-endpoint review-loop contract plus `/api/score`, persists all cycle state, degrades each model head independently to a deterministic fallback so the service never hard-fails, and serves the built SPA — a single URL is the whole product. A terminal review path shares the same backend route and renders the official Track-2 review format directly, with scraper-artifact cleanup, template-validation retries, and originality judgments grounded in reviewer summaries — verified end-to-end on real papers.

6 Web Application

The web client (React SPA over the contract-v2 API) exposes: paper submission (text/PDF); OpenReview-style review posts with rating/confidence/facet chips; a messenger-style rebuttal thread with per-comment replies; hunk-level revision approval in the manuscript pane; live agent-progress streams; a finalize view with meta-review, decision post, tiered score gauge (poster/spotlight/oral/notable-top-5%), and evidence-grounded attributions; multi-cycle resubmission history; and an analysis view placing the paper’s score trajectory on the real 47k-submission corpus distribution.

7 Artifacts

All artifacts are private pending a redistribution-rights review of OpenReview text. Models (Hugging Face, LoRA adapters with cards): `ac-metareview-v1/-v2/-v21`. Datasets: `ac-metareview-sft-v2/-sft-v21` (training pairs), `ac-metareview-score-data` (+ a 374-cluster criticism dictionary with per-type reject rates), `astage-reviewer-sft/-sft-v2` (paper $\rightarrow$ review, v2 with the reject fix), `icml-ac-threads-long` (discussion threads), and a 9,598-paper virtual meta-review corpus for ICML accepts — usable as author feedback, explicitly *not* as training data per the collapse results. Live

endpoints: a GCE backend serving the twelve-endpoint review-loop contract and `/api/score`; VESSL model serving with `/meta-review` and `/score`.

8 Limitations and Future Work

The paper $\rightarrow$ review stage currently uses frontier-LLM reviewers: our first fine-tuned reviewer was rejected for an accept-only training distribution (zero quality discrimination, rating-vs-label AUC 0.48), and the reject-augmented retrain (3,500 rejected-paper pairs, year-stripped prompts, trained on 4 $\times$ H100) also failed its gates — harsh ratings stayed near zero and rating-vs-label discrimination stayed random, indicating that severity calibration cannot come from review SFT alone (reviewers disagree on the same paper); the next path pairs a label-supervised rating regressor with the FT reviewer’s text. Author-only rebuttals do not move the judge’s decision — correctly mirroring real ACs discounting unverified claims — so demonstrating rebuttal value requires the planned reviewer-reaction simulator, for which training pairs exist. Novelty judgments are time-anchored to the base model’s knowledge; the inference-side fix (retrieving related recent work as evidence, distinct from the rejected review-retrieval) is designed but unmeasured. Score fine-ranking within accepted papers (the regime an author who will be accepted actually cares about) is the weakest measured capability. Feedback attribution currently grounds criticisms in verbatim manuscript evidence; a DeCAF-style analysis of the judge’s representation — facet concept probes over judge embeddings (24.9k labeled forums) with score-head gradient alignment — has its infrastructure prepared but no measured results yet. All evaluation above the decision layer relies on LLM judges; a blind human comparison is pending, as is a redistribution-rights review of OpenReview text (all datasets remain private).

9 Conclusion

A review agent is only as trustworthy as its measurements. The system presented here beats its strongest cheap baseline where it matters (borderline papers, +10pt), produces feedback that aligns with what authors go on to change *better than the real AC’s* (0.70 vs. 0.65), and wraps the whole pipeline — reviewers, rebuttal dynamics, meta-review, calibrated scoring — in the same process structure as the venue it simulates. That accuracy was not obtained in one shot: it is the product of six measure-reject-correct cycles, whose single surviving principle — review-agent behavior lives in the training distribution, not the prompt — also sets the direction of every next improvement. The full pipeline — data, training, evaluation, serving — re-runs in about three hours on four H100s, putting the next iteration hours away, not weeks.